MATH 342 Notes

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1 First Order Vector-Valued ODE

1.1 Repeated eigenvalues

1. Example: Find a fundamental set for $\dot{x} = \begin{bmatrix} 1 & -1 \\ 4 & 5 \end{bmatrix} x$.

$$\lambda = 3, \quad \xi = [1, -2]^T \tag{1}$$
$$(d/dt - \lambda)^2 x = 0 \text{ has solns } Ce^{\lambda t},\ Cte^{\lambda t} \tag{2}$$
$$\tag{3}$$

Propose

$$x = te^{\lambda t}\xi\,? \tag{4}$$

but then

$$\dot{x} - Ax = e^{\lambda t}\xi \neq 0 \tag{5}$$

so x is not a solution.

However, we can cancel off the nonhomogeneous part with an ansatz solution $e^{\lambda t}\eta$, $\eta \in \mathbb{R}^2$, to the nonhomogenous equation $\dot{x} - Ax = e^{\lambda t}\xi$. (Compare with method of undetermined coefficients.)

So

$$x = te^{\lambda t}\xi - e^{\lambda t}\eta. \tag{6}$$

Then $\dot{x} - Ax = 0$ gives

$$0 = e^{\lambda t}\xi + (\lambda e^{\lambda t}\eta - Ae^{\lambda t}\eta), \tag{7}$$

and therefore

$$e^{\lambda t}(\xi + \lambda\eta - A\eta) = 0 \implies \xi + \lambda\eta - A\eta = 0 \tag{8}$$
$$\implies \xi = (A - \lambda I)\eta. \tag{9}$$

Notice that $A - \lambda I$ is not invertible. But concretely we want to solve

$$\begin{bmatrix} -2 & -1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \tag{10}$$

Is ξ always in the column space of $A - \lambda I$?

2. Summary (choice of a fundamental set).

   In general when A has repeated eigenvalues, and A is 2×2,

$$x^1 = e^{\lambda t}\xi, \quad x^2 = te^{\lambda t}\xi + e^{\lambda t}\eta, \tag{11}$$

   where η is a solution to $(A - \lambda I)\eta = \xi$.

   η is called a *generalised eigenvector.*

   Highly important result. If η is *any* vector independent of ξ, we will have $(A - \lambda I)\eta = c\xi$ for some $c \neq 0$. (Why?)

1.2 Nonhomogeneous system

1. The system of interest is

$$\dot{x} = P(t)x + g(t), \qquad P(t) \in M_{n\times n}(\mathbb{R}), \quad x(t), g(t) \in M_{n\times 1}(\mathbb{R}) \tag{12}$$

   Solution is given as a sum of a particular solution with a general solution of the corresponding homogeneous system (which we have already done). We will focus on finding particular solutions x_p to (12).

2. Methods:

   (a) Undetermined coefficients for special $g(t)$

   (b) Laplace transforms

   (c) Decoupled systems for diagonalizable A

   (d) **Variation of parameters** (we will use this one)

   Example: $\dot{x} = \begin{bmatrix} 1 & -1 \\ 4 & 5 \end{bmatrix} x + \begin{bmatrix} e^t \\ 0 \end{bmatrix}$.

3. Proposed solutions (VOP):

$$x(t) = u_1(t)x^1(t) + u_2(t)x^2(t) \qquad \text{(undetermined coefficients)} \tag{13}$$

   where u_1, u_2 are scalar valued and x^1,x^2 solve the homogeneous equation. The goal is to find u_1, u_2.

   (a) Rewrite

$$x(t) = \begin{bmatrix} x^1(t) & x^2(t) \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \tag{14}$$

$$= \Psi(t)u(t) \tag{15}$$

$$= \begin{bmatrix} e^t & e^{5t} \\ -3e^t & e^{5t} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \tag{16}$$

   (b) Apply $x(t) = \Psi(t)u(t)$ to $\dot{x}(t) = Ax(t) + g(t)$ to get

$$\Psi'(t)u(t) + \Psi(t)u'(t) = A\Psi(t)u(t) + g(t) \tag{17}$$

   (c) Remember $\Psi(t)$ is invertible, and also explicitly

$$\Psi(t) = \begin{bmatrix} x^1(t) & x^2(t) \end{bmatrix} \tag{18}$$

$$\Psi'(t) = \begin{bmatrix} Ax^1(t) & Ax^2(t) \end{bmatrix} \tag{19}$$

$$= A\Psi(t). \tag{20}$$

Ψ solves a "matrix version" of the original homogeneous equation. We use this result to simplify (17):

$$\Psi(t)u'(t) = g(t). \tag{21}$$

(d) We have

$$\begin{bmatrix} e^t & e^{5t} \\ -3e^t & e^{5t} \end{bmatrix} \begin{bmatrix} u_1'(t) & u_2'(t) \end{bmatrix} = \begin{bmatrix} e^t & 0 \end{bmatrix}, \tag{22}$$

so

$$\begin{cases} e^t u_1'(t) + e^{5t} u_2'(t) = e^t \\ -3e^t u_1'(t) + e^{5t} u_2'(t) = 0 \end{cases} \tag{23}$$

A solution is

$$\begin{cases} u_1'(t) = 1/4 \\ u_2'(t) = \frac{3}{4} e^{-4t} \end{cases} \tag{24}$$

(e) Recover $u(t)$ from $u'(t)$.

$$u(t) = \int u'(t)dt = \int \begin{bmatrix} \frac{1}{4} \\ \frac{3}{4} e^{-4t} \end{bmatrix} dt = \begin{bmatrix} \frac{1}{4} t \\ -\frac{3}{16} e^{-4t} \end{bmatrix} + c \tag{25}$$

(f) One particular solution is

$$x(t) = \left(\int_0^t \frac{1}{4} ds \right) \begin{bmatrix} e^t \\ -3e^t \end{bmatrix} + \left(\int_0^t \frac{3}{4} e^{-4s} ds \right) \begin{bmatrix} e^{5t} \\ e^{5t} \end{bmatrix}. \tag{26}$$

4. *Solution method in general.*

Problem

$$\dot{x}(t) = P(t)x(t) + g(t).$$

Corresponding homogeneous system

$$\dot{x}(t) = P(t)x(t),$$

solved by

$$\dot{x}_h = P(t)x_h.$$

Solution to the nonhomogeneous system

$$x(t) = \Psi(t)u(t)$$

(solve for $u(t)$).

Equation for $u(t)$

$$\Psi'(t)u(t) + \Psi(t)u'(t) = P(t)\Psi(t)u(t) + g(t).$$

OBSERVE that $\Psi'(t) = P(t)\Psi(t)$ (even in the nonconstant $P(t)$ case), so that

$$\Psi(t)u(t) = g(t).$$

Furthermore, Ψ *is invertible.* This is because of the whole Wronskian nonzero $\implies$ nonzero everywhere / diffeq's do not collapse the dimension of subspaces spanned by solutions thing. Thus

$$u(t) = \int \Psi(t)^{-1} g(t) dt \tag{27}$$

$$\implies x_p(t) = \Psi(t) \left(c + \int_0^t \Psi(s)^{-1} g(s) ds \right) \tag{28}$$

$$= \Psi(t) c + \int_0^t \Psi(t) \Psi(s)^{-1} g(s) ds, \tag{29}$$

(which is even the general solution to the original equation, with degree of freedom c to be determined by initial conditions).

Special case: $P(t) = A$ (constant). Then

$$x(t) = e^{tA} x(0) = \int_0^t e^{(t-s)A} g(s) ds \qquad \text{(Duhamel's formula)} \tag{30}$$

2 Power Series Solutions to ODE

2.1 Basic power series tools

1. Example.

$$f(x) = \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} \frac{2^n}{n} x^n. \tag{31}$$

(a) Find the radius of convergence.

The radius of convergence of a power series $f(x) = \sum_{n=1}^{\infty} a_n (x - x_0)^n$ is a number $R \geq 0$ such that

i. $\forall x$ satisfying $|x - x_0| < R$, $f(x)$ converges absolutely (as in, $\sum_{n=1}^{\infty} |a_n| |x - x_0|^n$ converges).

ii. $\sum_{n=1}^{\infty} a_n (x - x_0)^n$ diverges whenever $|x - x_0| > R$.

R is called the *radius of convergence.*

(Here $R = 1/2$ by the Ratio Test.)

Tools for finding the radius of convergence

The Root Test: If $\lim_{n \to \infty} |a_n|^{1/n}$ converges, then $R = \dfrac{1}{\lim_{n \to \infty} |a_n|^{1/n}}$.

For instance, (31) gives

$$R = \frac{1}{\lim_{n \to \infty} (2^n/n)^{1/n}} = \frac{1}{\lim_{n \to \infty} 2/n^{1/n}} = 1/2 \qquad \text{since } n^{1/n} \to 1 \tag{32}$$

The Ratio Test: If $\lim_{n \to \infty} \dfrac{|a_{n+1}|}{|a_n|}$ converges, then $R = \dfrac{1}{\lim_{n \to \infty} \frac{|a_{n+1}|}{|a_n|}}$.

In our example,

$$R = \frac{1}{\lim_{n \to \infty} \frac{2^{n+1}/(n+1)}{2^n/n}} = \frac{1}{\lim_{n \to \infty} 2n/n + 1} = 1/2. \tag{33}$$

(b) Find $f'(x)$ (for meaningful x)

As usual, differentiate term by term. If $f(x)$ is given by a power series $\sum_{n=0}^{\infty} a_n x^n$, then

$$f'(x) = \sum_{n=0}^{\infty} n a_n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1} \tag{34}$$

By making a change of variable $k = n - 1$, and noticing that the first term drops out of the above series (because $n = 0$), we have

$$f'(x) = \sum_{k=0}^{\infty} (k+1) a_{k+1} x^k. \tag{35}$$

This procedure for getting the power series into the desired powers of x is called *index shifting*. It is important to keep track of the initial value for k and n in the sum. This is analagous to changing the bounds of integration when performing a change of variable.

A remarkable feature of the radius of convergence is that it is *invariant under differentiation or integration of the power series*. Consider our example. Differentiating term by term or using (35), the derivative series is

$$f'(x) = \sum_{k=0}^{\infty} (k+1) \frac{2^{k+1}}{k+1} x^k = \sum_{n=0}^{\infty} 2^{n+1} x^n. \tag{36}$$

By the ratio test, the radius of convergence of $f'(x)$ is

$$R = \frac{1}{\lim_{n\to\infty} \frac{2^{n+2}}{2^{n+1}}} = \frac{1}{2}, \tag{37}$$

which is the same as the radius of convergence for $f(x)$. The radius of convergence of the integral power series $\int dx f(x)$ is a fun exercise (it should also be 1/2).

(c) Some notions. Let

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \qquad \text{with} \qquad R > 0. \tag{38}$$

i. $f(x)$ is *analytic* at $x = x_0$. (This is what it means to be analytic at x_0; there exists a convergent power series equal to the function on a neighborhood of x_0.)

ii. $a_n = f^{(n)}(x_0)/n!$, and $f(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \ldots$, and in particular $f(x_0) = a_0$.

iii. Taylor series expansion: When f is analytic at x_0, it has a particular power series representation for $f(x)$, given by

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \tag{39}$$

For example, the exponential function is its own derivative and $e^0 = 1$. This is enough to define the exponential everywhere by its Taylor series at $x_0 = 0$:

$$e^x = \sum_{n=0}^{\infty} \frac{(e^x)^{(n)}\big|_{x=0}}{n!} x^n = \sum_{n=0}^{\infty} \frac{e^0}{n!} x^n = \sum_{n=0}^{\infty} \frac{1}{n!} x^n, \tag{40}$$

which should look familiar.

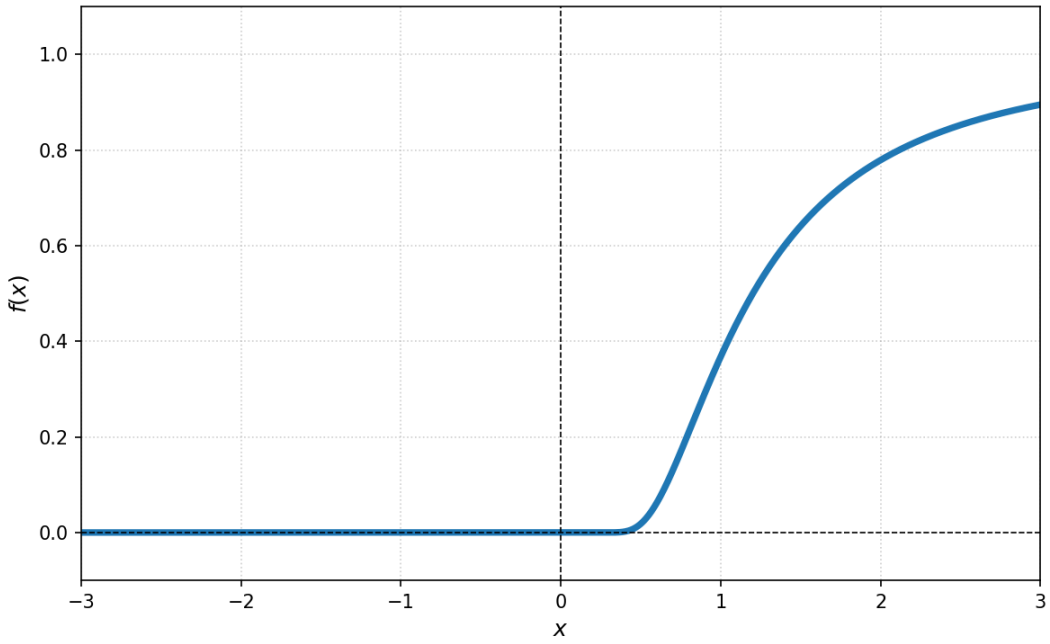
2. Analytic vs infinitely differentiable functions.

   Question: Every function that is analytic at $x = 0$ is infinitely differentiable at $x = 0$. Is the converse true? NO!

   Example: A classic example is the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases} \tag{41}$$

   This function is infinitely differentiable at $x_0 = 0$ (why?) but is not analytic there. For if it were analytic, its Taylor series would be identically zero since $f^{(n)}(0) = 0$ for all n (look at the value of f for negative x arbitrarily close to zero).



   Fact: $\sum_{n=0}^{\infty} a_n x^n = 0 \implies a_n = 0$ for all n. This is just a restatement of what we said above. (If you define $f(x)$ by that power series then we identify the a_n with the derivatives of f, which are all zero since f is by assumption identically zero.)

2.2 Homogeneous DE with variable coefficients

Consider the general type of ordinary differential equation

$$P(x)y'' + Q(x)y' + R(x)y = 0 \qquad (y = y(x) \text{ is real valued}) \tag{42}$$

We will make the following assumption.

Assumption: P, Q, R are polynomials and there is no scalar c such that $P(x), Q(x), R(x)$ are all divisible by $(x - c)$ (they are essentially "coprime" as elements of the ring of polynomials over the real or complex numbers).

1. Well known equations.

   (a) Airy's equation:

$$y'' - xy = 0, \qquad P(x) = 1, Q = 0, R(x) = -x. \tag{43}$$

(b) Bessel's equation:

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0, \qquad P(x) = x^2, Q(x) = x, R(x) = x^2 - \nu^2 \tag{44}$$

where ν (the Greek letter "nu") is a parameter in the equation.

If $\nu \neq 0$, then our assumption (about the coprime nature of P, Q, R) is satisfied, but not if $\nu = 0$. In the case $\nu = 0$, the equation simplifies a fair bit.

(c) Legendre's equation:

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0, \tag{45}$$

where α is a parameter.

2. Solution methods.

Let us restate our DE in **standard form**.

$$y'' + \frac{Q}{P}(x)y' + \frac{R}{P}(x)y = 0 \tag{46}$$

or

$$y'' + p(x)y' + q(x)y = 0, \tag{47}$$

where $p(x) = \frac{Q}{P}(x)$, and $q(x) = \frac{R}{P}(x)$.

But wait! We need to verify that it is okay to divide by $P(x)$ (what if we are dividing by zero?). If $P(x_0) \neq 0$, then $P(x) \neq 0$ for all x near x_0, and the standard form is valid near x_0. In this situation, x_0 is called an **ordinary point**.

3. Example: Airy's equation.

$$y'' - xy = 0 \qquad (x \in \mathbb{R}) \tag{48}$$

We can do some quantitative analysis on Airy's equation and make some rough sketches of various solution curves to (48). This essentially boils down to using the equation to guess the sign and magnitude of y'' at different points on the sketch, and deducing the shape of the solution by forcing it to be consistent with this data. Unfortunately I can't draw in LaTeX, so refer to the course notes for details.

To proceed with the solution we use a power series ansatz

$$y(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n = \sum_{n=0}^{\infty} a_n x^n \tag{49}$$

where $x_0 = 0$.

Solving the (48) amounts to determining the coefficients a_n. To do this we need to plug $y(x)$ into (48). Let's calculate the derivatives (see the short section on power series):

$$y'(x) = \sum_{n=0}^{\infty} \frac{d}{dx}\left[a_n x^n\right] \tag{50}$$

$$= \sum_{n=1}^{\infty} n a_n x^{n-1} \tag{51}$$

$$y''(x) = \sum_{n=1}^{\infty} \frac{d}{dx}\left[n a_n x^{n-1}\right] \tag{52}$$

$$= \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}. \tag{53}$$

The lower limit of the sums is *very important*. We drop terms when they are zero so that we don't end up with x to negative powers in the derivative (e.g. $na_n = 0$ when $n = 0$).

Plugging into the differential equation, we have

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x\sum_{n=0}^{\infty} a_n x^n = 0 \tag{54}$$

$$\implies \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^{n+1} = 0 \tag{55}$$

Doing some index shifting with $k = n - 2$ in the first sum and $k = n + 1$ in the second sum, we get

$$\sum_{k=0}^{\infty}(k+2)(k+1)a_{k+2}x^k - \sum_{k=1}^{\infty} a_{k-1}x^k = 0. \tag{56}$$

Writing out the first few terms, you can convince yourself that we have

$$(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \dots) - (a_0x + a_1x^2 + a_2x^3 + a_3x^4 + \dots) = 0. \tag{57}$$

Now we can simply collect terms with the same powers of x and require that all of the coefficients are zero. We can do this because 1, x, x^2, x^3 ... are linearly independent functions on an *interval* (they are naturally called the Taylor basis for polynomials and power series). Since we know that the resulting power series is zero for x in an interval (this technically comes from our hypothesis that the ansatz $y(x)$ is a solution), we can use this to determine a_n for all n.

We could use (57) to determine $a_0, a_1, a_2, \dots$, but we would never find them all. So let's do it formally. From (56), we have

$$2a_2 + \sum_{k=1}^{\infty} \left[a_{k+2}(k+2)(k+1) - a_{k-1}\right] x^k = 0, \tag{58}$$

which gives us the equations

$$\begin{cases} 2a_2 = 0, \\ a_{k+2}(k+2)(k+1) - a_{k-1} = 0 \quad \forall k \geq 1 \end{cases} \tag{59}$$

The first equation gives $a_2 = 0$, and the second can be arranged to be

$$a_{k+2} = -\frac{a_{k-1}}{(k+2)(k+1)} \quad \forall k \geq 1 \tag{60}$$

$$\implies a_{k+3} = -\frac{a_k}{(k+3)(k+2)} \quad \forall k \geq 1. \tag{61}$$

So any coefficient determines the coefficient three terms later in the series $f(x)$. Once we have a_0, we can generate a_3, and then a_6, and so on. Naturally, the coefficients form three decoupled sequencies or "chains"

$$a_0, a_3, a_6, a_9, \dots, a_{3n}, \dots \tag{62}$$

$$a_1, a_4, a_7, a_{10}, \dots, a_{3n+1}, \dots \tag{63}$$

$$a_2, a_5, a_8, a_{11}, \dots, a_{3n+2}, \cdots = 0 \tag{64}$$

The last of these is constant zero because $a_2 = 0$. The other two are controlled by two free parameters a_0 and a_1. By computing a few of these a_ns you will quickly realise a pattern, which you can write down as an explicit formula for coefficients in the first two chains:

$$a_{3n} = \frac{a_0}{\prod_{k=1}^{n}(3k-1)(3k)} \tag{65}$$

$$a_{3n+1} = \frac{a_1}{\prod_{k=1}^{n} 3k(3k+1)} \tag{66}$$

(recall the notation $\prod_{k=1}^{n} q_k = q_1 \cdot q_2 \cdots q_n$).

With this information we can simplify our ansatz and obtain the general solution.

$$y(x) = \sum_{n=0}^{\infty} a_n x^n \tag{67}$$

$$= \sum_{n=0}^{\infty} a_{3n} x^{3n} + \sum_{n=0}^{\infty} a_{3n+1} x^{3n+1} \tag{68}$$

$$= \sum_{n=0}^{\infty} \frac{a_0}{\prod_{k=1}^{n}(3k-1)(3k)} x^{3n} + \sum_{n=0}^{\infty} \frac{a_1}{\prod_{k=1}^{n} 3k(3k+1)} x^{3n+1} \tag{69}$$

$$= a_0 y_1(x) + a_1 y_2(x), \tag{70}$$

where $y_1(x)$ and $y_2(x)$ are the two linearly independent solutions given by the power series in the previous line. This completes the derivation of the general solution to Airy's equation.

Radius of convergence Remember that power series functions are only defined within their radius of convergence. Let's check where our solution to the Airy's equation is actually defined!

For $y_1(x)$,

$$b_n = \frac{a_0}{\prod_{k=1}^{n}(3k-1)(3k)} \tag{71}$$

and so by the Ratio test

$$R = \frac{1}{\lim_{n\to\infty} \frac{|b_{n+1}|}{|b_n|}} \tag{72}$$

$$= \frac{1}{\lim_{n\to\infty} \frac{\prod_{k=1}^{n}(3k-1)(3k)}{\prod_{k=1}^{n+1}(3k-1)(3k)}} \tag{73}$$

$$= \frac{1}{\lim_{n\to\infty} \frac{1}{(3n-1)(3n)}} \tag{74}$$

$$= \infty. \tag{75}$$

Similarly, the radius of convergence of y_2 is also infinity. This means our power series converges everywhere and we have a solution on the whole real line.

As a final remark, we can imagine expanding the solution around some other point $x_0 \neq 0$. If we do this, we get faster convergence around the new x_0, which is important for numerical applications. In addition, we can imagine situations in which $R < \infty$, in which case it would be necessary to find power series solutions around a collection of points; these solutions can in some cases be "patched together" to get a solution on a larger domain than just $[-R, R]$.

4. Theorem: Consider the second order homogeneous ODE with variable coeffients

$$P(x)y'' + Q(x)y' + R(x)y = 0 \tag{76}$$

with standard form

$$y'' + p(x)y' + q(x)y = 0. \tag{77}$$

We suppose x_0 is an ordinary point ($P(x) \neq 0$), and $p(x) = Q(x)/P(x)$, $q(x) = R(x)/P(x)$, and furthermore that p and q are analytic at x_0.

Then

(a) There exist power series solutions y_1 and y_2 such that $y = C_1y_1 + C_2y_2$ for all solutions y defined at x_0;

(b) $\{y_1, y_2\}$ is a fundamental set;

(c) $R_j > 0$, where R_j is the radius of of convergence y_j around x_0;

(d) $R_1, R_2 \geq \min\{|z - x_0| : z \in \mathbb{C},\ z \text{ are zeros of } P\}$.

For example, in Airy's equation P has no zeros. By part (d) of this theorem, that means $R_1, R_2 \geq \min \varnothing = \infty$, so y_1 and y_2 have infinite radius of convergence (as we found in the previous section).

Note that to apply part (d), we have to search for zeros z (where $P(z) = 0$) *in the complex plane.*

Example 1: Consider the DE

$$(1 - x)y'' - y' = 0, \qquad x_0 = 0 \tag{78}$$

Then $P(x) = 1 - x$, $Q(x) = -1$, $R(x) = 0$. $P(x)$ has a single zero at $x = 1$, so the theorem says a fundamental set of power series solutions can be found that converge on the disk of radius 1 around 0. Let's verify this by solving the differential equation.

$$\begin{aligned} v = y' &\implies (1 - x)v' - v = 0 && (79)\\ &\implies \frac{dv}{v} = \frac{dx}{1 - x} && (80)\\ &\implies \ln|v| = -\ln|1 - x| + C && (81)\\ &\implies y' = v = \frac{C_1}{1 - x}, && (82) \end{aligned}$$

so

$$\begin{aligned} y(x) &= \int \frac{C_1}{1 - x} dx && (83)\\ &= -C_1 \ln(1 - x) + C_2 && (84)\\ &= C_1y_1(x) + C_2y_2(x) && (85) \end{aligned}$$

where

$$\begin{cases} y_1 = -\ln(1 - x) = \sum_{n=1}^{\infty} \frac{x^n}{n}, \\ y_2 = 1. \end{cases} \tag{86}$$

By the ratio test, $R_1 = 1$ (the radius of convergence of $-\ln(1 - x)$), and $R_2 = \infty$ because y_2 is constant. So both radii of convergence are ≥ 1, as predicted.

Example 2: Consider

$$(1 + 3x^2)y'' - xy' + 4y = 0. \tag{87}$$

$P(x) \neq 0$ when $x = 0$, which motivates taking $x_0 = 0$ again. We use the theorem to get a lower bound on the radius of convergence of power series solutions around x_0.

First we find the zeros of $P(x) = 1 + 3x^2$. Setting $P(z) = 0$ implies $z^2 = -1/3$, so the zeros of P are $z = \pm i/\sqrt{3}$. The theorem tells us that we can find power series solutions

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \tag{88}$$

with radius of convergence $R \geq \min |z - x_0| = \min\{|i/\sqrt{3} - 0|, | - i/\sqrt{3} - 0|\} = 1/\sqrt{3}$. Remember that this means the series converges for all $|x - x_0| < R$, i.e. when the *strict* inequality holds. The radius of convergence says nothing about what happens when $|x - x_0| = R$ (whether the series converges absolutely, or whether it converges at all).

Example 3: We have a differential equation

$$y'' + xy' + 2y = 0 \tag{89}$$

with *initial conditions*

$$y(0) = 1 \qquad y'(0) = -2. \tag{90}$$

The theorem will give us a trick to determine some useful information about the solutions without finding them explicitly.

Suppose $\phi(x)$ is a solution. Our task is to find $\phi''(0)$ and $\phi'''(0)$, because this will determine the second and third coefficients a_2, a_3 in the power series solution, and we already have the first two coefficients (by Taylor's formula (why?) the initial conditions give us $a_0 = 1$ and $a_1 = -2$).

Let's see how this bootstrapping process to find the coefficients of ϕ works. First, we isolate the highest order term to get $\phi''(0)$ (which will give us a_2).

$$y'' = -xy' - 2y \tag{91}$$

$$\implies \phi''(0) = -0\phi'(0) - \phi(0) = -1. \tag{92}$$

With this information in hand, we can do the same for $\phi'''(0)$:

$$y''' = -y' - xy'' - 2y' = -xy'' - 3y' \tag{93}$$

$$\implies \phi'''(0) = -0\phi''(0) - 3\phi'(0) = 0 - (3)(-2) = 6. \tag{94}$$

In this special example, we did not need $\phi''(0)$ to determine $\phi'''(0)$, but you will notice that in general we may need the lower order derivatives to get the next one. This is why this process is sometimes known as bootstrapping.

5. Euler's equation (aka the Cauchy-Euler equation), is given for constants a_0, a_1, a_2 by

$$a_2 x^2 \frac{d^2y}{dx^2} + a_1 x^1 \frac{dy}{dx} + a_0 x^0 y = 0, \quad (a_2 \neq 0) \qquad \implies P(x) = a_2 x^2, \quad P(0) = 0. \tag{95}$$

This equation is an example of a general situation that we are going to study. This situation is when we have a second-order linear homogeneous ODE

$$P(x)y'' + Q(x)y' + R(x)y = 0 \tag{96}$$

where x_0 is a **singular point** (i.e. it is not an ordinary point, $P(x_0) = 0$), but it is a **regular** singular point.

By dividing by $a_2 x^2$, we can assume without loss of generality that we are dealing with the equation

$$\frac{d^2y}{dx^2} + \frac{a_1}{x}\frac{dy}{dx} + \frac{a_0}{x^2} y = 0 \tag{97}$$

in standard form (both equations have the same solutions).

6. Classification of singular points.

Regular singular point

We say x_0 is a **regular singular point** of a differential equation in standard form ($y'' + p(x)y' + q(x)y = 0$) if both limits

$$\lim_{x\to x_0}(x-x_0)p(x), \quad \lim_{x\to x_0}(x-x_0)^2 q(x) \tag{98}$$

exist in $\mathbb{R}$ (in other words, they are finite). This definition may remind you of the notion of a *removable singular point* from complex function theory.

For example, in Euler's equation we have $p(x) = a_1/x$ and $q(x) = a_0/x^2$, $x_0 = 0$. In this case $\lim_{x\to x_0}(x-x_0)p(x) = a_1$ and $\lim_{x\to x_0}(x-x_0)^2 q(x) = a_0$.

However, Euler's equation can be easily solved with methods from second year calculus. Our first real motivation for studying singular points will come from **Bessel's equation**. This is the equation

$$y'' + \frac{1}{x}y' + \frac{x^2-\nu^2}{x^2}y = 0 \tag{99}$$

where ν is a parameter. Bessel's equation also has a regular singular point at zero, and is much less tractable than Euler's equation with our current tools.

Irregular singular point

If either of the limits above does not exist or blows up to ∞ or $-\infty$, then x_0 is called an **irregular singular point**.

Example: Solve the Euler equation

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = 0 \tag{100}$$

The idea behind Euler equations is the following observation. Notice that if we set $x = e^t$, then

$$x\frac{dy}{dx} = \frac{dx}{dt}\frac{dy}{dx} = \frac{dy}{dt}. \tag{101}$$

This is a great simplification of the linear term in our differential equation! But what about the second order term? With a little more calculus, we find

$$x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt} \tag{102}$$

This is fanastic, because it turns Euler's equation into a constant coefficient equation, which we know how to solve:

$$\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right) - 2\frac{dy}{dt} - 4y = 0. \tag{103}$$

This is the motivation for setting $x = e^t$.

The rest of the solution is familiar; we find the characteristic polynomial $\tilde{p}(z) = z^2 - 3z - 4 = (z-4)(z+1)$, with roots $z = 4$ and $z = -1$. So we can write the general solution

$$y = C_1e^{4t} + C_2e^{-t}. \tag{104}$$

But the original equation was not written in terms of t. Substituting back in $x = e^t$, we obtain the final solution in terms of x:

$$y(x) = C_1x^4 + C_2x^{-1}. \tag{105}$$

7. More about singular points.

We will do a few examples on how to find and classify singular points.

Example: Find the singular points of the following equation and classify them.

$$2x^2(1-x)y'' + 3xy' + (5-x)y = 0 \tag{106}$$

First we write down the coefficient polynomial $P(x) = 2x^2(1-x)$, and divide out to get

$$y'' + \frac{3x}{2x^2(1-x)}y' + \frac{5-x}{2x^2(1-x)}y = 0 \tag{107}$$

in standard form. The singular points are the zeros of P, which are $x_0 = 0$ and $x_0 = 1$.

For $x_0 = 0$,

$$\lim_{x\to x_0}(x-x_0)p(x) = \lim_{x\to 0} x\frac{3x}{2x^2(1-x)} = \frac{3}{2} \in \mathbb{R} \tag{108}$$

$$\lim_{x\to x_0}(x-x_0)^2 q(x) = \lim_{x\to 0} x^2\frac{5-x}{2x^2(1-x)} = \frac{5}{2} \in \mathbb{R} \tag{109}$$

both exist, so $x_0 = 0$ is a regular singular point.

For $x_0 = 1$,

$$\lim_{x\to x_0}(x-x_0)p(x) = \lim_{x\to 1}(x-1)\frac{3x}{2x^2(1-x)} = \lim_{x\to 1}\frac{-3}{2x} = -\frac{3}{2} \in \mathbb{R} \tag{110}$$

$$\lim_{x\to x_0}(x-x_0)^2 q(x) = \lim_{x\to 1}(x-1)^2\frac{5-x}{2x^2(1-x)} = \lim_{x\to 1}(1-x)\frac{5-x}{2x^2} = 0 \in \mathbb{R} \tag{111}$$

so $x_0 = 1$ is also a regular singular point.

Setting for the general case:

$$y'' + p(x)y' + q(x)y = 0, \qquad x \neq x_0, \tag{112}$$

where x_0 is a regular singular point.

We can do a clever manipulation (due to Frobenius) to make this equation look like Euler's equation. Multiplying by x^2, we get

$$x^2y'' + x[xp(x)]y' + [x^2q(x)]y = 0. \tag{113}$$

We have suggestively factored the terms so that the coefficients of y and y' are exactly the functions we know the limit of as $x \to x_0$. (Look at the definition of a regular singular point). Since the limits exist, these coefficients

$$xp(x), \qquad x^2q(x) \tag{114}$$

are analytic functions, and they have power series

$$xp(x) = p_0 + p_1x + p_2x^2 + p_3x^3 + \ldots \tag{115}$$

$$x^2q(x) = q_0 + q_1x + q_2x^2 + q_3x^3 + \ldots \tag{116}$$

If we let $\hat{p}(x) = xp(x)$ and $\hat{q}(x) = x^2q(x)$, then our differential equation becomes

$$x^2y'' + x\hat{p}(x)y' + \hat{q}(x)y = 0. \tag{117}$$

When x is very close to x_0, $\hat{p}(x) \approx p_0$ and $\hat{q}(x) \approx q_0$, and we see that this equation is very nearly the same as the Euler equation

$$x^2y'' + xp_0y' + q_0y = 0, \tag{118}$$

known as the **Euler counterpart** to the original equation.

So what can we do with this? Since the differential equations are similar, we might imagine their solutions are also similar (and we already know the solutions to the Euler's equation). Assuming no repeated roots, the solutions to Euler's equation are just linear combinations of monomials. This line of thinking eventually leads to an ansatz for our original equation.

Frobenius ansatz: Solutions to (112) (assuming no repeated roots in the associated characteristic polynomial) are of the form

$$y = x^r \quad \text{plus "something"} \tag{119}$$

2.3 The method of Frobenius

In this section we will make a synthesis of the theory of the previous section to present a general solution strategy for homogeneous ODE with polynomial coefficients.

1. On the Euler counterpart and the Frobenius ansatz.

   The Euler counterpart of (112) is

   $$x^2y'' + xp_0y' + q_0y = 0, \tag{120}$$

   where $xp(x) = p_0 + p_1x + p_2x^2 + \dots$ and $x^2q(x) = q_0 + q_1x + q_2x^2 + \dots$

   Recall the general solution strategy for Euler's equation is making the substitution $x = e^t$, where one recovers the solutions $e^{rt}(= x^r)$, r being a root of the characteristic polynomial of the constant coefficient equation in t that Euler's equation reduces to. If characteristic polynomial has repeated roots, the second solution is given by $te^{rt}(= (\ln|x|)x^r)$. We will use this to formulate the Frobenius ansatz.

   Idea: Let's just work with the x^r solution (to the Euler counterpart) for now. The $(\ln|x|)x^r$ solution will come into play later.

   We come up with a candidate solution

   $$y(x) = x^r\left(\sum_{n=0}^{\infty} a_nx^n\right) = \sum_{n=0}^{\infty} a_nx^{n+r}, \tag{121}$$

   which takes the approximate solution x^r and perturbes it by an analytic function that we solve for. Note that (121) is not in general a power series, because r might not be an integer (or it might be negative)!

   Example: Solve

   $$2x^2y'' + 3xy' - (x+1)y = 0. \tag{122}$$

   In standard form, the equation is

   $$y'' + \frac{3x}{2x^2}y' - \frac{x+1}{2x^2}y = 0. \tag{123}$$

   Is zero a regular point? Checking the two limits,

   $$\lim_{x\to 0} xp(x) = \lim_{x\to 0} x\frac{3x}{2x^2} \quad \text{exists} \tag{124}$$

   $$\lim_{x\to 0} x^2q(x) = \lim_{x\to 0} x^2\frac{x+1}{2x^2} \quad \text{exists} \tag{125}$$

   so $x_0 = 0$ is a regular singular point. We need that it is regular for the Frobenius ansatz to apply.

Our ansatz is

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+r}. \tag{126}$$

We may assume $a_0 \neq 0$, for otherwise we could just shift r by one (r and a_k are undetermined coefficients), and get

$$y(x) = \sum_{n=1}^{\infty} a_n x^{n+r} \tag{127}$$

$$= \sum_{n=0}^{\infty} a_{n+1} x^{n+1+r} \tag{128}$$

$$= \sum_{n=0}^{\infty} b_n x^{n+1+r}. \tag{129}$$

You may then say "well, why can't $b_0 = 0$?" If $b_0 = 0$, then we repeat this process again and you might complain "but now we could have $c_0 = 0$!". But if we repeat this process for long enough, eventually one of the coefficients must be nonzero, otherwise they would all be zero and we would have the absurdity $y \equiv 0$!

With that out of the way, we can assume $a_0 \neq 0$ and solve for all the coefficients. First of all, we have

$$y(x) = \sum_{n=0}^{\infty} a_n x^{r+n} \qquad (a_0 \neq 0) \tag{130}$$

$$\implies y'(x) = \sum_{n=0}^{\infty} a_n (r_n) x^{r+n-1} \tag{131}$$

$$\implies y''(x) = \sum_{n=0}^{\infty} a_n (r+n)(r+n-1) x^{r+n-2} \tag{132}$$

Applying these formulas to (122), we have

$$0 = 2x^2 \sum_{n=0}^{\infty} a_n (r+n)(r+n-1) x^{r+n-2} + 3x \sum_{n=0}^{\infty} a_n (r+n) x^{r+n-1} - (x+1) \sum_{n=0}^{\infty} a_n x^{r+n} \tag{133}$$

$$= \sum_{n=0}^{\infty} a_n (r+n)(r+n-1) 2x^{r+n} + \sum_{n=0}^{\infty} a_n (r+n) 3x^{r+n} - \sum_{n=0}^{\infty} a_n x^{r+n+1} - \sum_{n=0}^{\infty} a_n x^{r+n+1}. \tag{134}$$

Rearranging and collecting terms (and using index shifting), we get

$$0 = \sum_{n=0}^{\infty} a_n \left(2(r+n)(r+n-1) + 3(r+n) - 1\right) x^{r+n} - \sum_{k=1}^{\infty} a_{k-1} x^{r+k} \tag{135}$$

$$= (2r(r-1) + 3r - 1) a_0 x^r + \sum_{n=1}^{\infty} \left[a_n \left(2(r+n)(r+n-1) + 3(r+n) - 1\right) - a_{n-1}\right] x^{r+n} \tag{136}$$

$$= (2r(r-1) + 3r - 1) a_0 x^r + \sum_{n=1}^{\infty} \left[\left(2(r+n)(r+n-1) + 3(r+n) - 1\right) a_n - a_{n-1}\right] x^{r+n} \tag{137}$$

$$= (2r^2 + r - 1) a_0 x^r + \sum_{n=1}^{\infty} \left[\left(2(r+n)^2 + (r+n) - 1\right) a_n - a_{n-1}\right] x^{r+n} \tag{138}$$

where we have used $k = n + 1$. This implies that all the coefficients of the x^m are zero, which gives us the recurrence equation

$$0 = \left(2(r+n)^2 + (r+n) - 1\right) a_n - a_{n-1} \tag{139}$$

$$\implies a_n = \frac{a_{n-1}}{2(r+n)^2 + (r+n) - 1} \tag{140}$$

as well as the equation from the x^r term:

$$2r^2 + r - 1 = 0 \tag{141}$$

since $a_0 \neq 0$. This is the famous **indicial equation** of Frobenius. Notice that a_0 is left as a free variable; it will become a coefficient in the general solution (remember we have not imposed any initial conditions thus far).

Remark The indicial equation (141) turns out to be precisely the *characteristic equation for the Euler counterpart*. This fact is a general feature of this technique of Frobenius.

2. Summary on obtaining the indicial equation.

   Start with the differential equation

$$P(x)y'' + Q(x)y' + R(x)y = 0 \tag{142}$$

   and formulate its Euler counterpart

$$x^2y'' + p_0xy' + q_0y = 0. \tag{143}$$

   Then apply the trial solution $y = x^r$ in the Euler counterpart, and you will get a polynomial equation in r. This is the indicial equation, and it corresponds to the characteristic equation of the Euler counterpart after substituting $x = e^t$ and rewriting it as a constant coefficient equation in terms of t.

   For instance, if the original equation is

$$2x^2y'' + 3xy' - (1+x)y = 0 \tag{144}$$

   as in our previous example, then the Euler counterpart is

$$2x^2y'' + 3xy' - y = 0 \tag{145}$$

   (factors of 2 don't matter). Trying $y(x) = x^r$ gives

$$(2r(r-1) + 3r - 1)x^r = 0 \tag{146}$$

   and so the indicial equation is

$$2r^2 + r - 1 = (2r-1)(r+1) = 0. \tag{147}$$

3. Using the indicial equation to find the general solution.

   Let's finish off our example. The roots of the indicial equation $r_1 = 1/2$, $r_2 = -1$ give us two possible solutions with the Frobenius ansatz, namely

$$y_1(x) = x^{1/2} \sum_{n=0}^{\infty} \alpha_n x^n \tag{148}$$

$$y_2(x) = x^{-1} \sum_{n=0}^{\infty} \beta_n x^n \tag{149}$$

Where α_n and β_n are found by plugging into the original differential equation, but by assumption $\alpha_0 \neq 0$ and $\beta_0 \neq 0$. I claim y_1 and y_2 are linearly independent.

To prove this, we need to show that the only solution to $c_1 y_1(x) + c_2 y_2(x) = 0$ that holds for all x is $c_1 = c_2 = 0$. But

$$c_1 y_1(x) + c_2 y_2(x) = 0 \tag{150}$$

$$\implies c_1 x^{1/2} \sum_{n=0}^{\infty} \alpha_n x^n + c_2 x^{-1} \sum_{n=0}^{\infty} \beta_n x^n = 0 \tag{151}$$

$$\implies c_1 x^{3/2} \sum_{n=0}^{\infty} \alpha_n x^n + c_2 \sum_{n=0}^{\infty} \beta_n x^n = 0. \tag{152}$$

Taking the limit as $x \to 0$ in the last equation, the only term that survives is $c_2 \beta_0$. Since $\beta_0 \neq 0$, we must have $c_2 = 0$. But then plugging $c_2 = 0$ back into the equation, we have

$$c_1 x^{3/2} \sum_{n=0}^{\infty} \alpha_n x^n = c_1(\alpha_0 x^{3/2} + \alpha_1 x^{5/2} + \dots) = 0 \tag{153}$$

As usual for power series, all of the terms of this series must be zero! (One way to think about it is that they are all linearly independent functions, so if a linear combination of them adds up to zero then all the coefficients must be zero.) Since $\alpha_0 \neq 0$, just looking at the first term of this series we can conclude that $c_1 \alpha_0 = 0 \implies c_1 = 0$ as well. We have been forced to conclude $c_1 = c_2 = 0$ as desired, so we are done. The two solutions we get from Frobenius are linearly independent.

If you look at the argument we just made, you will see that in the general case, in place of (152) we will get

$$c_1 x^{r_1 - r_2} \sum_{n=0}^{\infty} \alpha_n x^n + c_2 \sum_{n=0}^{\infty} \beta_n x^n = 0. \tag{154}$$

(by our choice of naming r_1 and r_2, we can assume without loss of generality that $r_1 - r_2 > 0$; the case $r_1 = r_2$, i.e. when the indicial equation has only one root, is another thing altogether that we will come back to in a future section). Again we send $x \to 0$ and obtain $c_2 = 0$ and consequently $c_1 = 0$. But here we have to be extremely careful about our assumptions. We assumed that the two Frobenius ansatz solutions y_1 and y_2 exist, *with the first coefficient nonzero.* It may be that the recurrence relations one derives to determine the coefficients of y_1 and y_2 only have solutions when the $\alpha_0 = 0$ or $\beta_0 = 0$, violating our hypothesis.

The moral of the story is that we have to settle for something weaker. The true result is that whenever $r_1 - r_2$ is *not an integer*, then y_1 and y_2 as constructed from the Frobenius ansatz will be linearly independent (even if the differential equation forces $\alpha_0 = 0$ or $\beta_0 = 0$). We will revisit this result soon.

4. Summary of The General Result

We started with the problem of obtaining solutions to

$$P(x)y'' + Q(x)y' + R(x)y = 0 \tag{155}$$

by using power series around a *regular singular point* $x_0 = 0$.

First we converted the equation into standard form and multiplied by x^2:

$$x^2 y'' + x(xp(x))y' + x^2 q(x)y = 0 \tag{156}$$

where $p(x) = Q(x)/P(x)$ and $q(x) = R(x)/P(x)$. We expanded $xp(x)$ and $x^2q(x)$ as power series around x_0:

$$xp(x) = p_0 + p_1x + p_2x^2 + \dots$$
$$x^2q(x) = q_0 + q_1x + q_2x^2 + \dots$$

which gives us an Euler equation that is kind of a first approximation to the original equation, called the Euler counterpart:

$$x^2y'' + p_0xy' + q_0y = 0. \tag{157}$$

Motivated by the form of solutions to the Euler counterpart, we introduced the Frobenius ansatz

$$y(x, r) = \sum_{n=0}^{\infty} a_n(r)x^{n+r}, \qquad a_0(r) \neq 0 \tag{158}$$

(we can even assume $a_0(r) = 1$). Plugging the ansatz into the differential equation gives a consistency equation for r, the indicial equation

$$F(r) := r(r-1) + p_0r + q_0 = 0 \tag{159}$$

($F(r)$ is called the **indicial polynomial**). The roots of the indicial polynomial r_1, r_2 index two linearly independent solutions

$$y_1(x) = y(x, r_1) = \sum_{n=0}^{\infty} a_n(r_1)x^{n+r_1}, \tag{160}$$

$$y_2(x) = y(x, r_2) = \sum_{n=0}^{\infty} a_n(r_2)x^{n+r_2} \tag{161}$$

provided $r_1 - r_2$ is not an integer. $a_n(r_1)$ and $a_n(r_2)$ are just the undetermined coefficients we called α_n and β_n earlier.

This is what we had. And it works when $F(r)$ has two distinct roots. But what if it only has one root? This is a problem we will extend our method to solve in the future. Before we do that, we will elaborate on the other special case where the above fails; when $r_1 - r_2$ is an integer.

5. Suppose $r_1 - r_2$ is an integer, say $r_1 - r_2 = -N \in \mathbb{Z}$. (The negative sign is for convenience later.)

   We show that y_1 and y_2 are not in general linearly independent. We have that $c_1y_1 + c_2y_2 = 0$ takes the following form:

$$c_1 \sum_{n=0}^{\infty} \alpha_n x^{r+n} + c_2 \sum_{n=N}^{\infty} \beta'_n x^{r+n} = 0 \tag{162}$$

   where we have set $r = r_1$, and $\beta'_n = \beta_{n-N}$. Assume $N \geq 0$. We can rewrite the above as

$$c_1 \sum_{n=0}^{N-1} \alpha_n x^{r+n} + \sum_{n=N}^{\infty} (c_1\alpha_n + c_2\beta'_n)x^{r+n} = 0 \tag{163}$$

   Now it is clear that as long as $\alpha_0 = \alpha_1 = \dots = \alpha_{N-1} = 0$, it can be the case that c_1 and c_2 are both nonzero and the α_n and β'_n terms simply cancel out. Since we cannot conclude $c_1 = c_2 = 0$, we cannot conclude that y_1 and y_2 are linearly independent (and in general they need not be).

   This failure to give linearly independent solutions does have a resolution, but it requires more advanced theory that we will not get into at the present moment. Remember, though, that at least y_1 and y_2 are linearly independent whenever $r_1 - r_2$ is not an integer. Hopefully you can see that in this case, the two sums cannot be "overlapped" as in (163). No power of x has coefficients in both y_1 and y_2, so the coefficients can never cancel each other out as we just imagined, and the solutions are therefore always linearly independent for $r_1 - r_2 \notin \mathbb{Z}$.